

CARBERRY, MB, Canada

WMO: 711700

Lat: 49.9058N

Lon: 99.3575W

Elev: 1258

StdP: 14.04

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-22.6	-33.5	0.9	-27.3	-28.1	1.2	-22.1	28.8	14.7	25.4	16.2	6.7	270	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	86.1	69.6	82.8	68.2	79.9	66.0	74.1	82.1	71.4	79.0	69.0	76.4	11.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	122.6	78.5	68.9	111.1	75.8	66.4	101.8	73.4	38.6	82.5	36.1	79.3	34.0	76.3	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.3	22.2	19.7	DB	-34.5	93.0	6.1	3.2	-38.9	95.3	-42.4	97.2	-45.8	99.0	-50.2	101.3	(4)
(5)			WB	-34.3	78.3	6.2	3.1	-38.7	80.5	-42.4	82.3	-45.8	84.0	-50.3	86.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	3.5	7.0	20.1	38.1	50.8	61.3	65.5	63.8	55.1	40.6	23.6	8.7		
	DBStd	24.84	15.36	14.24	14.48	11.19	8.80	6.80	5.70	6.13	8.14	8.93	12.23	14.28		
	HDD50	6483	1442	1205	929	377	98	4	0	0	40	314	793	1281		
	HDD65	10540	1907	1625	1393	808	445	148	64	96	309	757	1242	1746		
	CDD50	1613	0	0	1	20	124	343	481	429	193	22	0	0		
	CDD65	195	0	0	0	0	6	37	80	60	12	0	0	0		
	CDH74	2515	0	0	0	7	154	445	844	826	228	11	0	0		
	CDH80	646	0	0	0	0	34	107	203	251	50	1	0	0		
	WSAvg	9.2	9.3	9.2	9.3	10.2	10.8	9.3	7.7	7.8	8.8	9.8	9.5	9.0		
	PrecAvg	20.7	0.8	0.7	1.0	1.2	3.2	3.6	2.9	2.5	1.5	1.5	0.7	0.9		
	PrecMax	28.6	1.8	2.2	3.0	2.3	6.7	5.7	4.6	6.0	3.0	3.5	2.6	1.7		
(15) Precipitation	PrecMin	16.4	0.2	0.0	0.0	0.2	0.4	1.6	0.6	0.4	0.3	0.1	0.1	0.2		
	PrecStd	3.4	0.4	0.6	0.8	0.7	1.4	1.2	1.2	1.2	0.7	0.8	0.6	0.4		
	DB	37.9	39.3	58.4	74.2	84.5	88.2	89.3	91.8	85.6	74.2	57.5	40.9			
	MCWB	34.7	35.6	48.0	54.9	61.6	72.0	74.4	72.0	65.7	58.4	47.7	36.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	34.6	35.1	48.5	67.8	78.3	82.7	85.0	86.3	80.2	66.0	49.5	35.6			
	2% MCWB	32.7	33.2	42.7	51.5	57.6	66.3	72.3	69.8	63.6	53.0	42.6	33.1			
	5% DB	30.6	32.4	42.0	62.7	73.2	79.0	81.8	82.2	75.8	60.1	43.5	31.3			
	5% MCWB	29.2	31.2	38.1	48.9	55.8	64.5	70.4	68.1	61.5	50.6	38.8	29.7			
	10% DB	26.1	28.3	37.6	56.8	68.1	75.5	78.9	78.5	70.6	55.0	39.0	27.9			
	10% MCWB	25.1	27.2	35.3	46.6	53.5	63.0	68.6	66.0	59.1	48.3	35.9	26.9			
	0.4% WB	35.1	36.3	49.3	57.3	65.2	75.8	78.9	76.7	70.9	59.1	48.2	37.0			
	0.4% MCDB	37.1	38.7	56.3	70.3	77.4	84.2	83.9	85.2	79.2	72.1	56.7	40.1			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	32.9	33.6	43.5	53.5	61.1	70.9	75.0	72.5	66.5	55.1	42.9	33.6			
	2% MCDB	34.8	35.0	49.1	65.3	71.3	77.5	82.0	81.9	75.1	62.0	48.0	35.8			
	5% WB	29.4	31.2	38.8	50.1	58.4	67.4	72.4	69.9	63.7	51.8	39.1	30.1			
	5% MCDB	30.7	32.5	42.2	60.5	67.8	74.5	79.6	78.7	72.3	59.0	43.0	31.5			
	10% WB	25.4	27.4	35.7	47.0	55.8	64.7	70.0	67.7	60.8	48.5	36.0	27.3			
	10% MCDB	26.4	28.5	38.1	56.1	65.5	72.5	76.7	76.0	68.9	54.6	38.6	28.3			
	MDBR	18.0	19.5	18.8	21.1	23.2	20.7	22.1	24.8	23.4	19.0	16.2	16.3			
	5% DB	19.5	18.3	20.5	31.6	32.4	26.8	26.2	30.6	31.6	29.2	23.1	18.9			
(36) Mean Daily Temperature Range	MCWBR	17.7	16.1	14.9	17.7	15.6	13.5	14.9	15.9	16.4	17.3	16.4	16.7			
	MCDBR	19.4	17.2	19.9	28.5	24.7	21.4	23.7	26.0	26.3	27.3	22.0	18.2			
	MCWBR	17.9	15.3	14.9	16.8	14.2	13.1	15.1	15.3	15.7	17.0	15.9	16.3			
	MCDBR	17.9	15.3	14.9	16.8	14.2	13.1	15.1	15.3	15.7	17.0	15.9	16.3			
(40) Clear-Sky Solar Irradiance	taub	0.252	0.262	0.291	0.344	0.360	0.369	0.381	0.379	0.342	0.293	0.264	0.236			
	taud	2.299	2.291	2.359	2.353	2.368	2.380	2.354	2.356	2.443	2.547	2.474	2.381			
	Ebn at Noon	256	283	293	286	285	282	276	271	271	271	253	249			
	Edh at Noon	21	28	31	36	37	36	37	35	29	22	18	17			
(44) All-Sky Solar Radiation	RadAvg	453	829	1247	1599	1761	1880	1993	1643	1192	718	398	324			
	RadStd	36	59	96	110	148	128	84	116	90	93	55	36			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.35	-1.96	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	-0.95	-1.31	N/A	N/A	N/A	N/A

Nomenclature: See separate page